

Feasible Operating Region for Distributed Flexibility: Aggregation, Dispatch & Fairness

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Also referenced: Mariana · Caroline

The meeting reviewed progress on the first journal article — the computation of the **Feasible Operating Region (FOR)** of a distribution feeder and its use for dispatching Dynamic Operating Envelopes (DOEs) across aggregators and prosumers. Discussion centred on paper structure, the FOR aggregation/disaggregation methodology, physical interpretation of the current results, and the planned case studies. The consensus was that the work is progressing well and is roughly **two weeks ahead** of the schedule agreed a month earlier.

1 Regulatory Limits & Modelling Conservatism

Prof. Luis opened by clarifying how strict the technical standards really are, to reassure the team that small limit excursions in the model are not fatal:

- **Voltage — lower bound:** compliance is required for ~95% of the time, measured in **15-minute blocks** (a ~5% allowance to go below).
- **Voltage — upper bound:** must hold 100% of the time, but the metric is the **15-minute average**. Brief excursions of a minute or two do not constitute a violation — only a sustained breach beyond 15 minutes does.
- **Currents (thermal):** conductors typically tolerate **~30% overload for about an hour**; a 1% excess is immaterial. Transformers behave similarly, with published time-vs-overload tables (e.g., 30% for ~1 h, 50% for ~½ h). Higher overloads (~40%) are permitted only briefly, in emergencies, thanks to thermal inertia.

Take-away: The current model is deliberately **conservative** — it keeps everything below 100% loading and voltages below 1.1 p.u. at *all* times. Reality is more lenient, so any marginal violation that appears when validating dispatches (e.g., in OpenDSS) can be reasonably defended against the actual standards.

2 Paper Status & Structure

- **Introduction** drafted by Nacho; **methodology** (FOR calculation) being written by Tulio. Still pending: the DOE dispatch logic and the fairness-based disaggregation.
- The paper already runs to ~**10–11 pages**, so space is at a premium.

Decision — remove the related-work comparison table. Rather than a "checkbox" table of prior papers (which would also require a reference per row), the contribution and research gap will be argued **in the text**: group the prior work, state what it fails to address, then close each paragraph with how this paper breaches the gap. This frees space for methodology and results. The table would only be retained if it added information the text cannot convey — which was not the case here.

Narrative strategy

Structure the paper to **grow in difficulty** so reviewers can follow it (maximising the chance of acceptance): begin with a small **toy network** to explain the method end-to-end, then apply it to the realistic **9-transformer** case. In the results, show **one representative transformer in detail** (its FOR across the 48 periods plus average state of charge) and then the **total FOR at the PCC** — rather than exhaustively plotting all nine, which would be repetitive. "Choose the transformer that looks prettiest."

3 Core Methodology — Aggregating the FOR to the PCC

The computational problem

The model is currently solved **transformer by transformer**, and the nine individual FORs are combined via a **Minkowski sum** to obtain the aggregate FOR at the Point of Common Coupling (PCC). Solving the full network in a single optimisation (CPLEX / AIMMS) succeeds only for the first period and then crashes. The Canadian HPC cluster that Tulio has access to does **not have AIMMS installed** (installation is being requested).

Recommended approach (Prof. Luis)

Complement — or replace — the pure Minkowski sum with an explicit **power flow on the medium-voltage (MV) network** in OpenDSS:

- Represent each secondary transformer as an **equivalent P/Q load**, taken from that transformer's FOR result at each period.
- Run it as a **time series** over the 48 periods — the nine load profiles come directly from the FORs. This is done **offline** (post-processing the optimisation output), so there is no need to couple OpenDSS with AIMMS at every solver step.

Advantages: it is **simpler to explain**, it captures network **losses** (making the aggregate more realistic than a plain sum), and it **verifies MV feasibility** directly.

Feasibility argument for the MV network: If every low-voltage network stays within its limits (each transformer below 100%), the MV network should remain inside its **design limits by construction** — the feeder was engineered to carry all transformers at full loading in the worst case. Therefore MV overloads are unlikely, and the power flow acts as a **validation check** rather than a binding constraint.

Key conceptual point: The PCC FOR must reflect **all LV networks operating at their maximum simultaneously** — not one transformer dispatching while the others sit idle. The present per-transformer FOR is computed assuming only that transformer's prosumers act, so the MV power-flow check is what confirms the combined, all-at-once operation stays safe.

Nacho raised an alternative: run an **OPF at the MV level** (transformers modelled as loads at their FOR points) to detect violations and, if needed, shrink each transformer's FOR slightly before the Minkowski sum. Luis agreed this is the correct fallback **if** the power flow reveals problems, but expects it will not be necessary given the LV-side safety margin. Plan: run the power flow first, then reconsider only if issues appear.

4 Interpretation of Current Results

4.1 Residual grid import while batteries still hold charge

Tulio investigated why the net exchange is not exactly zero even when batteries retain energy. Systematically relaxing constraints showed:

- Removing the **end-of-day state-of-charge (SOC)** constraint → no change.
- Relaxing the **thermal limits** (altering conductor resistance) → no change.
- Lowering the **minimum SOC (10%)** → the import decreased.

Explanation: the batteries deplete down to their 10% floor, and the small residual import exists to keep them at that minimum. The batteries in the dataset are small/legacy units (~3.6 kW / 5 kWh, from Luis's PhD ~10 years ago); modern units are larger (~5 kW / 10–11 kWh). Because a uniform size was applied to every house, on average some houses need slightly more energy than assumed.

Guidance (Luis): No need to rerun everything. When a result like this appears, the task is to *explain* it, not re-engineer the model to hide it — "that's the reviewer's question, and now we have the answer." Frame it in the discussion (uniform/undersized batteries, averaging across heterogeneous houses). Optionally update the battery size to a modern value for realism, but this is not mandatory for the paper.

4.2 Shape and meaning of the FOR

- At **night** the FOR sits entirely on the import side (does not cross zero) — physically correct: no PV, low charge, so the site imports. At **noon** it shifts to the export side, because batteries are near full and surplus PV flows to the grid.
- The FOR should be presented to the TSO/DSO as the **controllable deltas** — how far the operating point can move from its current position. That controllable range is the **battery flexibility**, which must be shown separately from the uncontrolled PV.
- **Two flexibilities exist:** PV (not controllable — inverters run at maximum power point) and batteries (controllable). The paper's case study assumes the **only controllable source is the batteries**.

Do not propose PV curtailment. It is controversial and a likely cause of rejection — reviewers will ask why free solar energy is curtailed instead of stored. The correct levers are charging the batteries or shifting demand. This assumption is easy to justify: PV should not be wasted, and demand is essentially unpredictable, whereas the aggregator/VPP genuinely controls battery charge/discharge.

4.3 Economic value & demand-side flexibility (for the Discussion)

Tulio noted that surplus PV feeding neighbouring customers and covering losses represents a flexibility the prosumer arguably should be paid for. Luis agreed this belongs in a **discussion paragraph** (it adds value) alongside the possibility of **demand response / time-of-use** — e.g., encouraging EV charging or appliance use during export peaks instead of curtailing PV. However, these should remain **extensions discussed in words only**: bringing them into the model would make it too large and blur the interpretation, since the clean message here is that **all flexibility comes from the batteries**.

4.4 Voltage-dependent loads

Providing the service pushes voltages toward the upper limit; with voltage-dependent (ZIP) loads, higher voltage means slightly higher demand, which offsets the PV surplus and marginally reduces the deliverable service volume. For now, keep loads modelled as **constant PQ**; the effect can be noted qualitatively.

5 Uncertainty & the Rolling Horizon

A postdoc had asked how uncertainty is handled, given that load and generation are deterministic ("discrete"). The team's position:

- The **rolling horizon** — re-optimising every 30 minutes — inherently manages uncertainty. Generation and demand change little over 30 minutes, and its influence over the remainder of the day progressively vanishes as the horizon is recomputed.
- This produces a **more accurate (and appropriately smaller/tighter) FOR**: unlike a single full-day optimisation with perfect foresight — which lets the grid "prepare" and overstate the deliverable service — the rolling horizon only offers what the system genuinely has at each step. Whether to operate this way is ultimately the TSO/DSO's decision.
- This point must be stated explicitly in the methodology (it also answers **Mariana's earlier comment on the stochastic aspect**): the wording should convey that the rolling horizon *gathers* the demand/generation uncertainty rather than eliminating it.

MPC question: The approach is **not** Model Predictive Control — that is a distinct control-theory framing and not relevant to how this work should be presented.

6 Case-Study Design (Nacho)

An important clarification: prosumer **locations are fixed** — this is an **operational** problem, not planning. Houses are physically wired to a given phase and cannot be reallocated in operation, and the current results already fix the prosumers in place, so they will be kept as-is.

The variable worth studying is therefore the **aggregator assignment**, not prosumer position. Luis and Nacho converged on the most compelling scenario:

Highlighted case: when one aggregator's prosumers are **electrically farther** from the transformer (a weaker position, more prone to constraint), enforcing **fairness** forces heavier curtailment across *all* participants — pulling the FOR into a more compact region. This is the key phenomenon to showcase.

Planned comparison:

- **Case A — concentrated:** aggregators grouped by electrical distance (one gets the near half of the feeder, another the far end). Results change markedly under fairness. Framed as more representative of **rural** networks.
- **Case B — evenly/normally distributed:** aggregators mixed everywhere (approximating the no-aggregator average result). Framed as more representative of **dense/urban** networks.

Note: the base FOR contains no fairness; imposing fairness necessarily makes the feasible set **more constrained/compact**. Nacho also observed that in practice neighbours often contract the same provider, lending realism to the concentrated case (Luis illustrated with Chile's former "duct law" that tied a housing development to a single utility provider).

7 Collaboration & Administrative

- Nacho has established a shared **GitHub repository** with agreed version-control practices; Tulio's recent development was tested on Nacho's machine and **reproduces identical results**.
- The project is ~**two weeks ahead** of the schedule set a month ago; Nacho will begin the multiple-aggregator implementation and AIMMS validation earlier than planned.
- **Authorship:** Prof. Luis is confirmed as a **co-author** (his name was inadvertently missing on the Overleaf and will be added). Tulio will also add **Mariana** to the Overleaf.
- Open channel: the team will use WhatsApp for anything urgent between meetings rather than waiting.

8 Action Items

Owner	Action	Timing
Tulio	On vacation next week (back 4 Aug). Implement the OpenDSS time-series power-flow check on the MV network; deepen the Minkowski-sum and fairness-based disaggregation; continue the methodology; consider updated battery sizing; add discussion paragraphs (economic value of PV surplus, DR extension); add Mariana to Overleaf; merge the GitHub branches.	Now → mid-Aug
Nacho	Review the methodology and provide comments; add references and smooth abrupt paragraph transitions; implement the multiple-aggregator logic; add Prof. Luis as co-author on Overleaf; develop the case-study scenarios (aggregator distribution by electrical distance).	Next week
Prof. Luis	Review the draft focusing on structure and direction — the introduction's storytelling and the ordering of the methodology — rather than a deep line-by-line review at this stage.	~14 Aug

Upcoming meetings

- **Tulio–Nacho one-to-one:** moved from 7 Aug to the week of ~11 Aug, giving Tulio time to act on comments after returning from vacation.
- **Full team meeting with Prof. Luis: Friday, 14 August** (same time).

9 Key Take-aways at a Glance

- Model conservatism is defensible — real standards allow short/averaged excursions.
- Drop the related-work table; argue the gap in text. Build the paper toy-network → realistic case; detail one transformer + the PCC total.
- Aggregate to the PCC via an **offline OpenDSS time-series power flow** (transformers as equivalent P/Q loads); LV-side safety implies MV feasibility, so this is a check, not a constraint. MV-OPF correction is the fallback only if needed.
- The end-of-day residual import is a **battery-sizing / minimum-SOC** effect — explain it, don't rerun.
- Present the FOR as the **controllable battery deltas**; never propose PV curtailment. Economic and demand-response angles go in the discussion only.
- The **rolling horizon** is the uncertainty mechanism (answers Mariana's stochastic comment); it is not MPC.
- Case studies vary **aggregator assignment**, not prosumer location; the headline result is fairness tightening the FOR when an aggregator is electrically distant.